Neuro Vision TFT (-W)

Datasheet



5.0 in 800 x 480 capacitive touch display expansion board for the Neuro N6, with a 5 MP auto-focus camera, a VL53L1CB Time-of-Flight ranging sensor, an illumination LED and a mono speaker amplifier. The TFT-W variant adds a u-blox MAYA-W476 module for 2.4 GHz Wi-Fi and BLE.

5.0 in
800 x 480 touch display

5 MP
OV5640 auto-focus camera

72°
Camera field of view

Wi-Fi/BLE
TFT-W variant

PART NUMBER	HARDWARE REVISION	DOCUMENT	DATE
OL-DEV-N6-006 / -007	Rev A / Rev B	OL-DEV-N6-006 Rev D	25 October 2026

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1 Overview

The Neuro Vision TFT is a display and camera expansion board for the Neuro N6. It combines a 5.0 in 800 x 480 capacitive touch display with an OmniVision OV5640 auto-focus camera, a Time-of-Flight ranging sensor, an illumination LED and a mono speaker amplifier, so that the Neuro N6 can show camera frames and inference results on the board and take input from the touch panel.

- 5.0 in TFT display, 800 x 480 pixels, 400 cd/m² typical luminance, 1000:1 contrast ratio, 80° viewing angle in all directions.
- RGB565 parallel interface driven by the Neuro N6 LTDC display controller.
- Capacitive touch panel with a GT911 controller: up to five touch points over I2C.
- Constant-current LED backlight from an on-board TPS61042 boost converter, switched with the display by the DISP signal.
- OmniVision OV5640 5 MP auto-focus camera (72° field of view) on a 2-lane MIPI CSI-2 interface: QSXGA at 15 fps, 1080p at 30 fps, 720p and VGA at 60 fps.
- VL53L1CB Time-of-Flight ranging sensor in long distance mode, ranging a 4 x 4 grid of zones.
- White illumination LED switched on and off from the camera STROBE output.
- MAX98357A mono class D amplifier with an on-board 8 Ω micro speaker, fed over I2S from the Neuro N6 SAI1.
- Reset button on the display side of the board.
- Powered from the Neuro N6 VIN, 3.3 V and 1.8 V rails, with three on-board LDO regulators for the camera and the Time-of-Flight sensor.
- One 30-pin and one 40-pin board-to-board connector, 3.5 mm mating height.
- TFT-W variant: u-blox MAYA-W476 module with 2.4 GHz Wi-Fi (802.11b/g/n) and BLE.
- Supported by the Neuro N6 Arduino core.

2 Introduction

The Neuro Vision TFT mounts on the Neuro N6 through two board-to-board connectors with a mating height of 3.5 mm. The Neuro N6 sits behind the display, and the two boards form a self-contained edge AI terminal: camera frames and inference results are shown on the display, the application is controlled from the touch panel, and the speaker gives audio feedback.

The board is available in two variants. The standard Neuro Vision TFT provides the display, touch, camera, depth sensing, illumination and audio functions. The Neuro Vision TFT-W adds a u-blox MAYA-W476 wireless module with 2.4 GHz Wi-Fi (802.11b/g/n) and BLE, for example to stream video frames or inference results over Wi-Fi. The TFT-W is offered only in countries where its assessed radio configuration is permitted.

The Neuro N6 Arduino core, available from neuron6.com, provides drivers and examples for the display, the touch panel, the camera, the Time-of-Flight sensor and the speaker. Unless stated otherwise, the information in this document applies to both variants.

For detailed information on the display module, the OV5640 image sensor, the VL53L1CB sensor, the MAX98357A amplifier and the MAYA-W476 module, refer to the OmniVision, STMicroelectronics, Analog Devices and u-blox documentation.

2.1 Variants and ordering information

VARIANT	PART NUMBER	HARDWARE REVISION	WIRELESS	DESCRIPTION
Neuro Vision TFT	`OL-DEV-N6-006`	Rev A	None	5.0 in touch display, 5 MP auto-focus camera, Time-of-Flight sensor, illumination LED and speaker
Neuro Vision TFT-W	`OL-DEV-N6-007`	Rev B	2.4 GHz Wi-Fi, BLE	Adds the u-blox MAYA-W476 module. Available in selected countries only

Table 1. Ordering information. Quote the part number and hardware revision when ordering and when contacting Ohm Lab for support.

2.2 Related products

- Neuro N6 (OL-DEV-N6-001)
- Neuro Vision OV5640 (OL-DEV-N6-002)
- Neuro Vision OV5640-W (OL-DEV-N6-003)
- Neuro Vision ST Cam (OL-DEV-N6-004)
- Neuro Vision Thermal (OL-DEV-N6-005)
- Neuro N6 Enclosure (OL-DEV-N6-009)
- Neuro N6 Debug Adapter (OL-DEV-N6-010)
- B-CAMS-IMX Adapter (OL-DEV-N6-011)

3 Features

FEATURE	DESCRIPTION
Display	
Display module	5.0 in normally black transmissive TFT LCD, 800 x 480 pixels in RGB vertical stripes, 108.00 x 64.80 mm active area, ST7262 driver, integrated capacitive touch panel
Optical characteristics	400 cd/m² typical luminance (350 cd/m² minimum) through the touch panel, 1000:1 contrast ratio, 80° viewing angle up, down, left and right, 30 ms response time
Display interface	16-bit RGB565 parallel data with DCLK, HSYNC, VSYNC and DE from the Neuro N6 LTDC; DISP display on/off control
Display operation	With the Arduino core: RGB565 camera layer with an ARGB4444 overlay; refresh follows the camera frame rate, 30 to 60 Hz; used for camera modes up to 800 x 480
Backlight	12 white LEDs, driven at a constant 40.6 mA in total by an on-board TPS61042 boost converter supplied from VIN; switched on and off with the display by DISP
Touch panel	Capacitive, GT911 controller, up to five touch points, I2C on the 3.3 V host bus; the Arduino core reads it by polling. Interrupt line SCREEN_INT routed to the Neuro N6
Camera	
OV5640 image sensor	5 MP CMOS, 2592 x 1944 (QSXGA), 1/4 in optical format, 1.4 x 1.4 µm pixels, rolling shutter with automatic exposure control, 2-lane MIPI CSI-2 output, SCCB control interface
Video modes	With the Arduino core: VGA 640 x 480 and WVGA 800 x 480 at 60 fps; XGA 1024 x 768 at 30 fps; 720p at 60 fps; 1080p (centre crop) at 30 fps; QSXGA 2592 x 1944 at 15 fps. Rates are capture maximums; lower rates can be set
Output format	RGB565 from the sensor; JPEG encoded on the Neuro N6 for streaming
Image controls	Automatic or manual exposure, automatic or manual gain (1 to 63.9x), automatic white balance, mirror and flip, through the Neuro N6 Arduino core
Lens module	Auto-focus lens with voice-coil motor, 72° field of view, 4-element plastic lens with IR filter, 1/4 in format, distortion below 1%; module 8.5 x 8.5 x 5.0 mm with a 24-pin FPC; single and continuous auto-focus by the OV5640 embedded controller, firmware downloaded over SCCB
Depth sensing	
VL53L1CB Time-of-Flight sensor	Absolute ranging independent of target colour and reflectance; long distance mode with a 4 x 4 grid of zones, one distance per zone; 940 nm Class 1 VCSEL; 27° field of view; interrupt routed to CONN_F (the Arduino core polls the sensor over I2C)
Illumination	
White LED	2835-package white LED switched by an SI2302S MOSFET from the OV5640 STROBE output; on or off under software control
Audio	
Amplifier	MAX98357A mono class D amplifier: I2S input from the Neuro N6 SAI1 block B, no master clock required, 6 dB gain, left and right channels mixed to mono, shutdown controlled by SPK_EN; 16-bit mono audio with the Arduino core
Speaker	On-board 16 x 9 x 3 mm micro speaker, 8 Ω, 0.7 W rated power

FEATURE	DESCRIPTION
Controls	
Reset button	Pulls NRST low to reset the Neuro N6 and the expansion board; two presses within 900 ms enter the Ohm Lab bootloader
Wireless connectivity (TFT-W only)	
u-blox MAYA-W476	NXP IW610: 2.4 GHz Wi-Fi 802.11b/g/n 1x1, 20 MHz channels, station or access point, WPA2 and WPA3; BLE peripheral and central; SDIO (Wi-Fi) and 4-wire HCI UART (BLE) host interfaces; embedded PCB antenna
Interfaces	
Board-to-board connectors	40-pin and 30-pin HC-PBB40C receptacles, 0.4 mm pitch, 3.5 mm mating height: RGB565 LTDC, 2-lane MIPI CSI-2, I2C (3.3 V host side), SAI1 I2S, touch and Time-of-Flight interrupts, display and amplifier enables, NRST, VIN, 3.3 V and 1.8 V; SDIO, HCI UART and wake signals on the TFT-W
FPC connectors	40-pin display, 6-pin touch and 24-pin camera FPC connectors, all 0.5 mm pitch
Power	
Supply	VIN, 3.3 V and 1.8 V from the Neuro N6. VIN feeds only the backlight converter, through a ferrite bead and with a separate backlight ground. Three on-board LDO regulators (2.8 V analog, 1.5 V core, 2.8 V Time-of-Flight)

Table 2. Neuro Vision TFT and TFT-W features.

4 Mechanical

The display is mounted on the front of the board, centred across the board width with its lower edge 0.2 mm below the lower board edge. Its display and touch FPCs fold round the lower board edge to their connectors on the rear. The camera module, the Time-of-Flight sensor, the illumination LED, the speaker and the reset button sit in the strip above the display; the camera FPC folds round the upper board edge to its connector on the rear. The connectors and the remaining circuitry are on the rear, where the Neuro N6 mounts back to back with the board on a 3.4 mm support. All dimensions are in millimetres and are taken from the PCB design data, the Neuro Vision TFT assembly model and the display module drawing.

PCB outline	133.00 x 87.00 mm
Overall outline	133.0 x 89.9 mm, including the display FPCs folded round the lower board edge (1.4 mm) and the camera FPC folded round the upper board edge (1.6 mm)
PCB	1.6 mm
Display module	120.70 x 75.80 x 4.85 mm including the touch panel; active area 108.00 x 64.80 mm
Height above the PCB (front)	5.0 mm to the display cover glass
Height below the PCB (rear)	3.4 mm (board-to-board receptacles)
Overall height	10.0 mm
Stack height on the Neuro N6	17.2 mm from the Neuro N6 battery connector to the display cover glass, with a 3.5 mm mating gap
Mounting holes	4 x Ø2.54 mm, 127.00 x 70.28 mm pattern, 3.00 mm from the side edges and the lower edge
Neuro N6 support	50.7 x 22.8 x 3.4 mm, between the boards under the Neuro N6, with openings for the board-to-board receptacles
Board-to-board receptacles	30-pin centred 19.81 mm and 40-pin centred 25.04 mm from the left board edge (front view), both 64.93 mm above the lower board edge

Table 3. Mechanical summary.

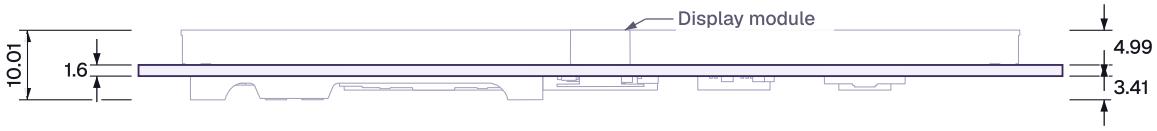


Figure 1. Neuro Vision TFT side view from the assembly model, with the display module, the camera module and the Neuro N6 support.

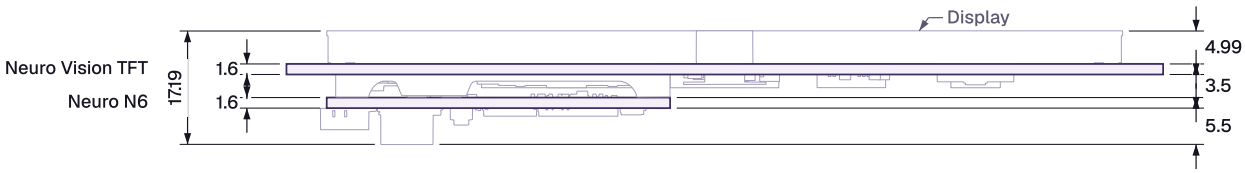


Figure 2. Neuro Vision TFT mounted on the Neuro N6, side view to scale from the two 3D models. The boards mount back to back with a 3.5 mm mating gap; the Neuro N6 component side faces away from the display. The Neuro N6 support fills the mating gap.

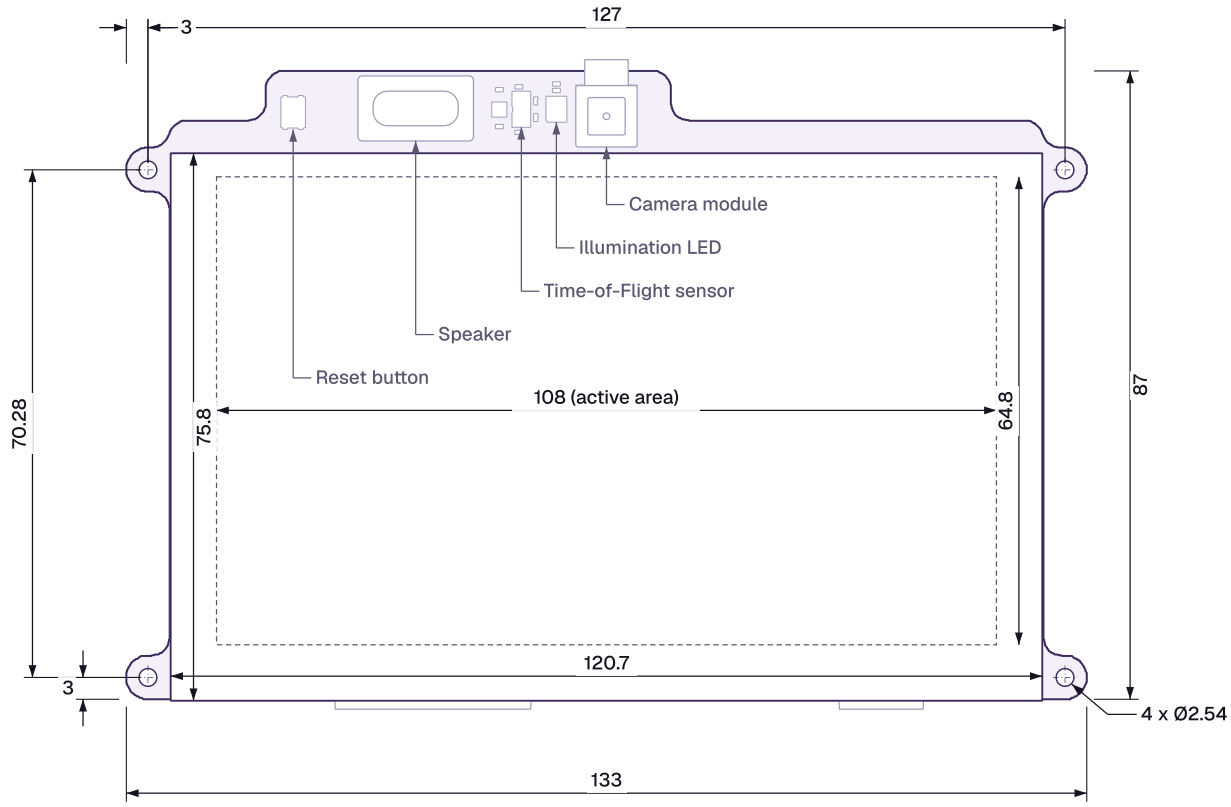


Figure 3. Neuro Vision TFT front view (display side). The board, the display module, the camera module and the speaker are taken from the assembly model and the active area (dashed) from the display module drawing. Dimensions are nominal design values in millimetres.

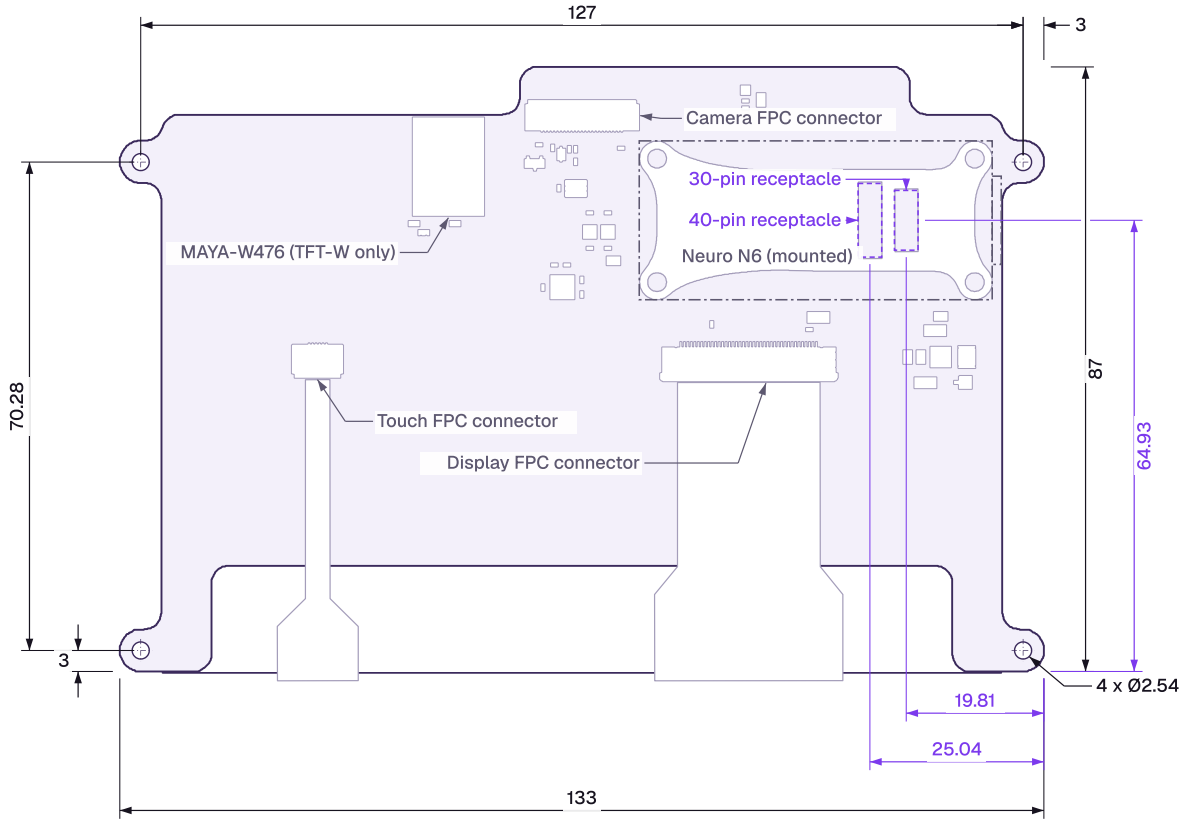


Figure 4. Neuro Vision TFT rear view, showing the board-to-board receptacles, the FPC connectors, the folded display and touch FPCs, the display seen through the cut-out, and the Neuro N6 support with the position of the Neuro N6 when mounted. The wireless module is fitted on the TFT-W only.

5 Block diagram

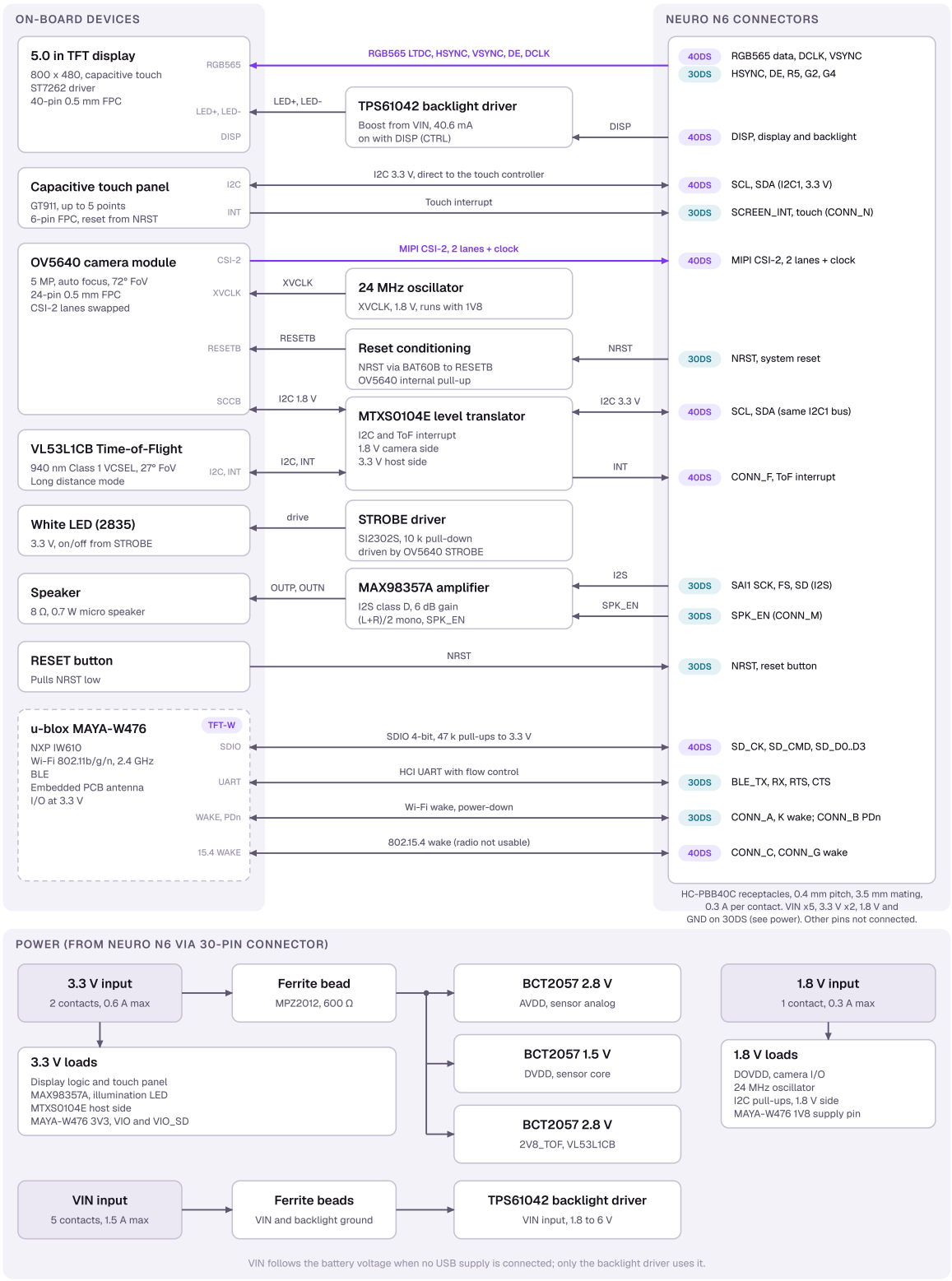


Figure 5. Neuro Vision TFT block diagram. The wireless module (dashed) is fitted on the TFT-W only.

6 Hardware interfaces

6.1 Display

The display is a 5.0 in normally black TFT LCD module with 800 x 480 pixels, an ST7262 driver and an integrated capacitive touch panel. It connects through a 40-pin, 0.5 mm pitch FPC connector and is driven directly by the Neuro N6 LTDC display controller with 16-bit RGB565 data, the pixel clock (DCLK), horizontal and vertical sync and data enable. The panel accepts 24-bit RGB data: the five most significant red and blue bits and the six most significant green bits are connected, and the remaining panel data inputs (R0 to R2, G0, G1 and B0 to B2) are not connected on this board.

DISP switches the display on and off. It is pulled up to 3.3 V through 10 kΩ, so the display is on unless the host drives DISP low, and it also drives the enable input of the backlight converter, so DISP switches the display and the backlight together. With the Arduino core the display shows camera modes up to 800 x 480 (VGA letterboxed, WVGA full screen), and its refresh rate follows the camera frame rate, from 30 to 60 Hz.

The backlight consists of 12 white LEDs with a forward voltage of 19.2 V typical at 40 mA in total. A TPS61042 boost converter supplied from VIN drives them at a constant current set by a 6.2 Ω sense resistor: the 252 mV feedback voltage gives 40.6 mA typical. The converter overvoltage protection limits its output to 29 V typical if the LEDs are disconnected. VIN reaches the converter through a ferrite bead, and the converter ground connects to the board ground through a second ferrite bead, which keeps the switching currents away from the camera and sensor circuits. The backlight current is fixed; the board has no separate brightness control.

> **Caution** The display glass is fragile. Do not press on the display surface or its edges, do not bend the display, touch or camera FPCs sharply, and do not connect or disconnect any FPC while the board is powered. The FPCs and the camera module are sensitive to electrostatic discharge; handle the board by its edges.

6.2 Touch panel

The capacitive touch panel is read by a Goodix GT911 controller on the display module, which reports up to five touch points; the Arduino core reads them by polling over I2C. It connects through a 6-pin, 0.5 mm pitch FPC connector and sits directly on the 3.3 V host I2C bus (Neuro N6 I2C1), with no level translation. The controller signals touch events on SCREEN_INT (30-pin connector pin 6, Neuro N6 CONN_N), and its reset line on the touch FPC connector is driven by the system NRST, so the controller is reset with the Neuro N6.

The GT911 selects its I2C address, 0x5D or 0x14, from the level of its INT pin when its reset is released. On this board the reset is NRST, and SCREEN_INT is not driven while the Neuro N6 is held in reset, so software should accept either address.

6.3 Camera module

The OV5640 camera module sits in the strip above the display and connects through a 24-pin, 0.5 mm pitch FPC connector on the rear of the board. The module contains the OV5640 sensor and a 72° auto-focus lens assembly driven by the sensor's integrated voice-coil motor controller; the auto-focus firmware is downloaded over the SCCB interface at start-up. An on-board 24 MHz oscillator drives XVCLK, and the sensor is reset by the system NRST signal (see the reset and clocking section). Image data leaves the board on the 2-lane MIPI CSI-2 interface routed to the 40-pin connector, with the data lanes in the reverse order to the Neuro Vision OV5640 (see the MIPI CSI-2 section).

6.4 Time-of-Flight sensor

A VL53L1CB ranging sensor is fitted next to the camera module. It emits invisible 940 nm light from an integrated Class 1 VCSEL and measures the photon return time on a SPAD array, giving absolute distance independent of target colour and reflectance over a 27° field of view. The Arduino core runs the sensor in long distance mode and ranges a 4 x 4 grid of zones, one distance per zone. Typical uses alongside the camera are presence detection and attaching absolute distance to detected objects. See the laser safety section for the classification of the emitter and the handling precautions.

The sensor shares the 1.8 V I2C bus with the camera. Its interrupt output (GPIO1) is level-translated to 3.3 V and routed to CONN_F (40-pin connector pin 26); the Arduino core polls the sensor over I2C and does not use the interrupt. XSHUT is pulled up to 1.8 V on the board, so the sensor is enabled whenever the board is powered.

6.5 Illumination LED

A high-brightness white LED next to the camera provides scene illumination. It is powered from the 3.3 V rail and switched by an SI2302S MOSFET driven from the OV5640 STROBE output; a 10 kΩ pull-down holds the LED off while the sensor is in reset. The Arduino core switches the LED on or off through the sensor's SCCB registers. No series resistor is fitted: the LED current is set by the LED forward voltage, the 3.3 V rail and the MOSFET on-resistance. Take this into account before leaving the LED on for long periods.

6.6 Audio

A MAX98357A mono class D amplifier drives the on-board speaker from the Neuro N6 SAI1 block B, used as an I2S transmitter. The amplifier needs only the bit clock, the frame clock and the data line; the SAI1 master clock is not connected. It runs from the 3.3 V rail, and its gain is fixed at 6 dB by connecting GAIN_SLOT to the supply. SPK_EN (Neuro N6 CONN_M) drives the SD_MODE input through 640 kΩ, which forms a divider with the 100 kΩ internal pull-down: with SPK_EN high at 3.3 V the amplifier plays the average of the left and right channels, (L+R)/2, and with SPK_EN low it is shut down.

The speaker is a Dragonstate HDK-160908ZA-12P, 16 x 9 x 3 mm, 8 Ω, rated at 0.7 W with 1.0 W maximum for one minute. Its resonant frequency is 550 Hz in free air, so reproduction below this frequency is limited. Set the digital signal level so that the speaker rating is not exceeded.

6.7 Reset button

The RESET button in the strip above the display pulls NRST low. It resets the Neuro N6 and the expansion board in the same way as the Neuro N6 reset button, which faces away from the display when the boards are stacked. Two presses within 900 ms enter the Ohm Lab bootloader (see the Neuro N6 datasheet).

6.8 Wireless module (TFT-W)

The TFT-W adds a u-blox MAYA-W476 module on the rear of the board, a host-based multiradio module built on the NXP IW610 with an embedded PCB antenna. It provides single-stream 2.4 GHz Wi-Fi (802.11b/g/n, 20 MHz channels) and BLE. With the Arduino core the module runs as a Wi-Fi station or access point with WPA2 or WPA3 security, and as a BLE peripheral or central. Wi-Fi traffic uses the Neuro N6 4-bit SDMMC1 interface (SDIO at 3.3 V, up to 50 MHz) and BLE uses the 4-wire USART6 HCI link, so Wi-Fi and BLE use separate host interfaces; the two radios share the 2.4 GHz antenna by time-sharing. The module takes its 3V3, VIO and VIO_SD supplies from the 3.3 V rail, so its I/O is referenced to 3.3 V, and its 1V8 supply from the 1.8 V rail.

Wake-up and control signals are routed to Neuro N6 GPIO: CONN_A and CONN_K for the Wi-Fi wake handshake, CONN_C and CONN_G for the 802.15.4 subsystem wake handshake, and CONN_B for the module power-down input (PDn). The module's 802.15.4 radio is present in silicon, but its SPI host interface is not connected, so it cannot be operated. The configuration pin CON[2] is pulled low with 51 kΩ; CON[0] and CON[1] are left open.

6.9 Connectors

CONNECTOR	PART NUMBER	MATING PART
Display FPC connector	`HC-FPC-0.5-40P-CSH20`	40-way, 0.5 mm pitch FPC (display module)
Touch FPC connector	`FPC-05F-6PH20`	6-way, 0.5 mm pitch FPC (touch panel)
Camera FPC connector	`FPC-05F-24PH20`	24-way, 0.5 mm pitch FPC (camera module tail)
30-pin board-to-board receptacle	`HC-PBB40C-30DS-0.4V-3.5-02`	`HC-PBB40C-30DP-0.4V-02` (Neuro N6)
40-pin board-to-board receptacle	`HC-PBB40C-40DS-0.4V-3.5-02`	`HC-PBB40C-40DP-0.4V-02` (Neuro N6)

Table 4. Connector part numbers.

The board-to-board receptacles mate with the plugs fitted to the underside of the Neuro N6 at a mating height of 3.5 mm. The two boards mount back to back, so the component sides of both boards face outwards. Align both connectors before pressing the boards together, and separate the boards evenly when removing the expansion board.

7 Board operation

7.1 Absolute maximum ratings

Absolute maximum ratings are stress ratings only. They are not operating conditions, and functional operation at or beyond these limits is neither implied nor guaranteed. Exposure to conditions at the limits may reduce reliability. Each limit is the lowest rating among the components connected to that net on the Neuro Vision TFT and TFT-W schematics. All voltages are referenced to GND.

PARAMETER	MIN	MAX	UNIT	LIMITING ELEMENT
Supply voltage, VIN	-0.3	7	V	TPS61042 VIN input
Supply voltage, VDD_3V3	-0.3	3.47	V	GT911 touch controller supply (display module)
Supply voltage, VDD_1V8 (TFT)	-0.3	3.6	V	VL53L1CB I/O (pull-ups to 1V8)
Supply voltage, VDD_1V8 (TFT-W)	-0.3	2.16	V	MAYA-W476 1V8 supply input
Voltage on SCL, SDA, SCREEN_INT and NRST	-0.3	3.47	V	GT911 I/O and reset input
Voltage on CONN_F	-0.3	VDD_3V3 + 0.3	V	MTXS0104E B port
Voltage on DISP	-0.3	VDD_3V3 + 0.3	V	Panel DISP input (3.3 V logic); also the TPS61042 CTRL input
Voltage on I2S inputs (SAI1_SCK_B, SAI1_FS_B, SAI1_SD_B)	-0.3	6	V	MAX98357A BCLK, LRCLK and DIN
Voltage on camera-domain 1.8 V signals (SCL1V8, SDA1V8, XCLK)	-0.3	VDD_1V8 + 0.3	V	MTXS0104E A port, VL53L1CB, oscillator
Voltage on SDIO, HCI UART and control pins (TFT-W)	-0.3	VIO + 0.4	V	MAYA-W476 input high level maximum (VIO = 3.3 V)
Continuous current per board-to-board contact		0.3	A	HC-PBB40C 0.4 mm pitch contact
Backlight forward current, per LED		25	mA	Display module backlight
Illumination LED forward current		60	mA	XL-2835UWC-02
Storage temperature	-30	80	°C	Display module

Table 5. Absolute maximum ratings.

VIN reaches the board on five contacts of the 30-pin connector, the 3.3 V rail on two and the 1.8 V rail on one, giving 1.5 A, 0.6 A and 0.3 A respectively at the 0.3 A per-contact limit.

7.2 Recommended operating conditions

PARAMETER	MIN	MAX	UNIT	CONDITIONS
Ambient temperature	0	30	°C	Indoor, dry locations only
Relative humidity	25	75	%	Non-condensing
Supply				From the Neuro N6, powered with 5 V DC via USB-C or its battery

Table 6. Recommended operating conditions.

7.3 Electrical characteristics

PARAMETER	MIN	TYP	MAX	UNIT	CONDITIONS
Supply voltage, VIN	3.5		5.5	V	From the Neuro N6: USB VBUS (5 V) or the battery voltage (3.5 to 4.2 V); the backlight converter operates from 1.8 V
Supply voltage, VDD_3V3	3.23	3.33	3.44	V	From the Neuro N6; the touch panel is specified to 3.3 V
Supply voltage, VDD_1V8	1.75	1.8	1.85	V	From the Neuro N6
Backlight LED current		40.6		mA	6.2 Ω sense resistor, 252 mV TPS61042 feedback voltage
Backlight LED voltage	18.0	19.2	20.4	V	At 40 mA in total
Backlight LED power		768		mW	At 40 mA; the converter input power is higher by the conversion losses
On-board analog rail, AVDD		2.8		V	BCT2057, sensor analog supply
On-board core rail, DVDD		1.5		V	BCT2057, sensor core supply
On-board Time-of-Flight rail, 2V8_TOF		2.8		V	BCT2057
Display input low voltage, VIL	0		0.99	V	0.3 x VDD_3V3 at 3.3 V; RGB, DCLK, HSYNC, VSYNC, DE, DISP
Display input high voltage, VIH	2.31		3.3	V	0.7 x VDD_3V3 at 3.3 V; RGB, DCLK, HSYNC, VSYNC, DE, DISP
Camera-domain logic level		1.8		V	DOVDD, I2C, XVCLK
Host-side logic level		3.3		V	LTDC, I2C, I2S, interrupts, SDIO, UART, control

Table 7. Electrical characteristics.

A Neuro N6 with an expansion board typically draws 150 to 500 mA from its 5 V USB supply, depending on the workload; see the Neuro N6 datasheet.

7.4 Power supply

The board has no power input of its own. It is powered from the Neuro N6 through the board-to-board connectors, which provide VIN and the 3.3 V (VDD_3V3) and 1.8 V (VDD_1V8) rails. VIN is the Neuro N6 rail after its ideal-diode OR-ing (USB VBUS or the battery), so it follows the battery voltage when no USB-C supply is connected. On this board VIN supplies only the backlight converter, through an MPZ2012 ferrite bead, and the converter ground returns to the board ground through a second ferrite bead.

The 3.3 V rail supplies the display logic and the touch panel, the audio amplifier, the illumination LED and the I2C level translator directly. Through a third ferrite bead it feeds three BCT2057 low-dropout regulators: one generates the 2.8 V analog supply (AVDD) for the image sensor on its own analog ground, joined to GND at one point, one generates the 1.5 V digital core supply (DVDD), and one generates an independent 2.8 V supply for the Time-of-Flight sensor and its VCSEL. The camera I/O domain (DOVDD), the 24 MHz oscillator and the 1.8 V pull-ups run directly from the 1.8 V rail. On the TFT-W the wireless module takes its 3V3, VIO and VIO_SD supplies from the 3.3 V rail and its 1V8 supply from the 1.8 V rail.

RAIL	SOURCE	VOLTAGE	SUPPLIES
`VIN`	Neuro N6, 30-pin connector	VBUS or VBAT	Backlight boost converter (via ferrite bead)
`VDD_3V3`	Neuro N6, 30-pin connector	3.3 V	Display logic and touch panel, audio amplifier, illumination LED, I2C level translator, LDO inputs (via ferrite bead), MAYA-W476 3V3, VIO and VIO_SD (TFT-W)
`VDD_1V8`	Neuro N6, 30-pin connector	1.8 V	Camera I/O (DOVDD), 24 MHz oscillator, 1.8 V I2C and sensor pull-ups, MAYA-W476 1V8 (TFT-W)
`AVDD`	On-board LDO	2.8 V	Camera analog supply; regulator on its own analog ground
`DVDD`	On-board LDO	1.5 V	Camera digital core
`2V8_TOF`	On-board LDO	2.8 V	Time-of-Flight sensor and VCSEL
`BLA`	TPS61042 boost converter	19.2 V typ	Backlight LEDs, 40.6 mA constant current in total

Table 8. Power rails.

> **Warning** Do not attach or detach the Neuro Vision TFT while the Neuro N6 is powered. Disconnect the USB-C supply and the battery before changing expansion boards; otherwise the boards may be permanently damaged.

7.5 Reset and clocking

The system reset signal NRST from the Neuro N6 resets the camera and the touch controller. The OV5640 active-low RESETB input is pulled low through a BAT60B diode whenever NRST is asserted and returns high through the sensor's internal pull-up when NRST is released. The touch controller reset line is connected to NRST directly. The RESET button on the board asserts NRST. The Time-of-Flight sensor is not connected to NRST; it is reset over I2C or by removing board power.

The 24 MHz oscillator runs whenever the 1.8 V rail is present and drives XVCLK directly, so no camera clock is required from the host. The display pixel clock is generated by the Neuro N6 LTDC; see the display interface section for the panel timing.

8 Peripheral interfaces

Connector pins are given as `40DS:n` and `30DS:n`, the receptacle pin numbers on this board. The matching Neuro N6 plug pin is listed in the connector tables, and the Neuro N6 signal name is given where it differs from the net name on this board. The Neuro N6 datasheet gives the MCU pin for every signal.

8.1 Display interface (LTDC)

The display is driven by the Neuro N6 LTDC in 16-bit RGB565 mode. All signals are 3.3 V logic, routed directly from the board-to-board connectors to the display FPC connector.

NET	CONNECTOR PIN	DESCRIPTION
`LCD_R7`	`40DS:4`	Panel R7, red data bit 7
`LCD_R6`	`40DS:6`	Panel R6, red data bit 6
`LCD_R5`	`30DS:4`	Panel R5, red data bit 5
`LCD_R4`	`40DS:36`	Panel R4, red data bit 4
`LCD_R3`	`40DS:2`	Panel R3, red data bit 3
`LCD_G7`	`40DS:32`	Panel G7, green data bit 7
`LCD_G6`	`40DS:34`	Panel G6, green data bit 6
`LCD_G5`	`40DS:28`	Panel G5, green data bit 5
`LCD_G4`	`30DS:14`	Panel G4, green data bit 4
`LCD_G3`	`40DS:30`	Panel G3, green data bit 3
`LCD_G2`	`30DS:26`	Panel G2, green data bit 2
`LCD_B7`	`40DS:24`	Panel B7, blue data bit 7
`LCD_B6`	`40DS:22`	Panel B6, blue data bit 6
`LCD_B5`	`40DS:16`	Panel B5, blue data bit 5
`LCD_B4`	`40DS:20`	Panel B4, blue data bit 4
`LCD_B3`	`40DS:18`	Panel B3, blue data bit 3
`LCD_CLK`	`40DS:14`	Panel DCLK, pixel clock
`LCD_HSYNC`	`30DS:1`	Panel HSYNC, horizontal sync
`LCD_VSYNC`	`40DS:10`	Panel VSYNC, vertical sync
`LCD_DE`	`30DS:11`	Panel DE, data enable
`DISP`	`40DS:40`	Display and backlight on/off, 10 kΩ pull-up to 3.3 V (Neuro N6 CONN_D)

Table 9. Display interface pins.

The Arduino core drives the panel with the timing below. The pixel clock is set from the camera frame rate, giving a panel refresh of 30 to 60 Hz.

PARAMETER	HORIZONTAL	VERTICAL
Sync pulse width	48 DCLK	3 lines
Back porch	88 DCLK	32 lines
Active	800 DCLK	480 lines
Front porch	40 DCLK	13 lines
Total	976 DCLK	528 lines

Table 10. Display timing set by the Arduino core. HSYNC, VSYNC and DE are active low; the pixel clock is 15.4 to 30.8 MHz.

8.2 MIPI CSI-2

The OV5640 2-lane MIPI CSI-2 interface is routed differentially from the camera FPC connector to the 40-pin connector, matching the Neuro N6 D-PHY receiver (CSI-2 version 1.3 compatible). Ground pins separate the lanes.

On this board the camera data lanes are connected in the reverse order to the Neuro Vision OV5640: camera data lane 0 reaches the Neuro N6 data lane 1 pins and camera data lane 1 the data lane 0 pins, with the polarity of each pair unchanged. Set the CSI-2 receiver data lane mapping to match when using this board.

NET	CONNECTOR PIN	NEURO N6 SIGNAL	DESCRIPTION
`CSI_D0P`	`40DS:15`	`CSI_D1_P`	Camera data lane 0 positive
`CSI_D0N`	`40DS:13`	`CSI_D1_N`	Camera data lane 0 negative
`CSI_CLKP`	`40DS:23`	`CSI_CLK_P`	Clock lane positive
`CSI_CLKN`	`40DS:21`	`CSI_CLK_N`	Clock lane negative
`CSI_D1P`	`40DS:31`	`CSI_D0_P`	Camera data lane 1 positive
`CSI_D1N`	`40DS:29`	`CSI_D0_N`	Camera data lane 1 negative

Table 11. MIPI CSI-2 pins.

8.3 I2C

A single I2C bus controls the touch controller, the camera and the Time-of-Flight sensor. It enters the board at 3.3 V on the 40-pin connector, from the Neuro N6 I2C1 bus that also serves the on-board IMU and magnetometer of the Neuro N6. The touch controller sits directly on the 3.3 V bus. An MTXS0104E translates the bus to the 1.8 V camera domain, where it is pulled up through 2.2 kΩ resistors. Standard mode (100 kHz) and fast mode (400 kHz) are supported.

NET	LOGIC LEVEL	CONNECTOR PIN	DESCRIPTION
`SCL`	3.3 V	`40DS:35`	I2C clock, host side and touch controller
`SDA`	3.3 V	`40DS:37`	I2C data, host side and touch controller
`SCL1V8`	1.8 V		I2C clock, camera domain
`SDA1V8`	1.8 V		I2C data, camera domain

Table 12. I2C pins.

DEVICE	7-BIT ADDRESS	NOTES
GT911 touch controller	`0x5D` or `0x14`	Selected when the controller leaves reset; see the touch panel section
OV5640 image sensor (SCCB)	`0x3C`	Write 0x78, read 0x79
VL53L1CB Time-of-Flight sensor	`0x29`	Default address, can be reprogrammed at run time

Table 13. I2C addresses.

8.4 I2S audio

The amplifier takes standard I2S data from the Neuro N6 SAI1 block B; no master clock is needed. The Arduino core sends 16-bit mono audio. The sample rate is derived from a 25 MHz clock, so the default 48 kHz request gives 48.8 kHz.

NET	MAX98357A PIN	CONNECTOR PIN	DESCRIPTION
`SAI1_SCK_B`	`BCLK`	`30DS:16`	I2S bit clock
`SAI1_FS_B`	`LRCLK`	`30DS:28`	I2S frame clock (left and right channel select)
`SAI1_SD_B`	`DIN`	`30DS:18`	I2S serial data
`SPK_EN`	`SD_MODE`	`30DS:7`	Amplifier enable through 640 kΩ: high for (L+R)/2 output, low for shutdown (Neuro N6 CONN_M)
`SAI1_MCLK_B`		`30DS:2`	SAI1 master clock, not connected on this board

Table 14. I2S audio pins.

8.5 Camera control signals

NET	OV5640 PIN	CONNECTOR PIN	DESCRIPTION
`XCLK`	`XVCLK`		24 MHz master clock from the on-board oscillator
`NRST`	`RESETB`	`30DS:29`	Active-low camera reset, driven by the system NRST through a BAT60B diode
`STROBE`	`STROBE`		Strobe output, switches the white LED

Table 15. Camera control signals.

8.6 Interrupt and control signals

NET	DEVICE PIN	CONNECTOR PIN	DESCRIPTION
`SCREEN_INT`	GT911 `INT`	`30DS:6`	Touch interrupt at 3.3 V (Neuro N6 CONN_N); the Arduino core polls the touch controller
`CONN_F`	VL53L1CB `GPIO1`	`40DS:26`	Ranging interrupt at 3.3 V, level translated; not used by the Arduino core, which polls the sensor
`DISP`	Panel `DISP` , TPS61042 `CTRL`	`40DS:40`	Display and backlight enable, active high, 10 kΩ pull-up to 3.3 V (Neuro N6 CONN_D)
`SPK_EN`	MAX98357A `SD_MODE`	`30DS:7`	Amplifier enable (Neuro N6 CONN_M)
	VL53L1CB `XSHUT`		Pulled up to 1.8 V on the board; the sensor is always enabled

Table 16. Interrupt and control signals.

8.7 SDIO (TFT-W)

On the TFT-W the Neuro N6 4-bit SDMMC1 interface connects to the module SDIO interface at 3.3 V, up to 50 MHz, and carries all Wi-Fi traffic. The command and data lines are pulled up to 3.3 V through 47 kΩ resistors. On the standard variant these pins are not connected.

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
SD_CK	SD_CLK (A4)	40DS:5	SDIO clock
SD_CMD	SD_CMD (B4)	40DS:3	SDIO command
SD_D0	SD_DAT[0] (B3)	40DS:8	SDIO data 0
SD_D1	SD_DAT[1] (A3)	40DS:1	SDIO data 1
SD_D2	SD_DAT[2] (B5)	40DS:39	SDIO data 2
SD_D3	SD_DAT[3] (A5)	40DS:7	SDIO data 3

Table 17. SDIO pins (TFT-W).

8.8 BLE HCI UART (TFT-W)

BLE traffic uses the Neuro N6 4-wire USART6 interface, connected to the module HCI UART. Signal names are given from the host's point of view.

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
BLE_TX	UART_SIN (G8)	30DS:3	Host transmit to module receive
BLE_RX	UART_SOUT (G9)	30DS:5	Module transmit to host receive
BLE_RTS	UART_CTS (H8)	30DS:9	Host request to send
BLE_CTS	UART_RTS (H9)	30DS:8	Host clear to send

Table 18. HCI UART pins (TFT-W).

8.9 Wireless control signals (TFT-W)

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
CONN_A	WL_WAKE_OUT / JTAG_TDI (F7)	30DS:30	Wi-Fi wake-up, module to host
CONN_K	WL_WAKE_IN (J8)	30DS:20	Wi-Fi wake-up, host to module
CONN_C	NB_WAKE_OUT / JTAG_TDO (E7)	40DS:38	802.15.4 subsystem wake-up, module to host
CONN_G	NB_WAKE_IN (J9)	40DS:12	802.15.4 subsystem wake-up, host to module
CONN_B	PDn (J3)	30DS:22	Module power-down input, active low

Table 19. Wireless control signals (TFT-W). CON[2] is pulled low on the board; CON[0] and CON[1] are open.

8.10 40-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	SD_D1	40	SDIO data 1 (TFT-W only)
2	LCD_R3	39	LTDC red data bit 3
3	SD_CMD	38	SDIO command (TFT-W only)

PIN	NET	N6 PLUG PIN	FUNCTION
4	LCD_R7	37	LTDC red data bit 7
5	SD_CK	36	SDIO clock (TFT-W only)
6	LCD_R6	35	LTDC red data bit 6
7	SD_D3	34	SDIO data 3 (TFT-W only)
8	SD_D0	33	SDIO data 0 (TFT-W only)
9	GND	32	Ground
10	LCD_VSYNC	31	LTDC vertical sync
11	GND	30	Ground
12	CONN_G	29	802.15.4 wake-up input (TFT-W only)
13	CSI_D0N	28	Camera data lane 0 negative (Neuro N6 CSI_D1_N)
14	LCD_CLK	27	LTDC pixel clock
15	CSI_D0P	26	Camera data lane 0 positive (Neuro N6 CSI_D1_P)
16	LCD_B5	25	LTDC blue data bit 5
17	GND	24	Ground
18	LCD_B3	23	LTDC blue data bit 3
19	GND	22	Ground
20	LCD_B4	21	LTDC blue data bit 4
21	CSI_CLKN	20	MIPI CSI-2 clock lane negative
22	LCD_B6	19	LTDC blue data bit 6
23	CSI_CLKP	18	MIPI CSI-2 clock lane positive
24	LCD_B7	17	LTDC blue data bit 7
25	GND	16	Ground
26	CONN_F	15	Time-of-Flight interrupt to the host
27	GND	14	Ground
28	LCD_G5	13	LTDC green data bit 5
29	CSI_D1N	12	Camera data lane 1 negative (Neuro N6 CSI_D0_N)
30	LCD_G3	11	LTDC green data bit 3
31	CSI_D1P	10	Camera data lane 1 positive (Neuro N6 CSI_D0_P)
32	LCD_G7	9	LTDC green data bit 7
33	GND	8	Ground
34	LCD_G6	7	LTDC green data bit 6
35	SCL	6	I2C clock, 3.3 V host side (Neuro N6 I2C1)
36	LCD_R4	5	LTDC red data bit 4
37	SDA	4	I2C data, 3.3 V host side (Neuro N6 I2C1)

PIN	NET	N6 PLUG PIN	FUNCTION
38	`CONN_C`	3	802.15.4 wake-up output (TFT-W only)
39	`SD_D2`	2	SDIO data 2 (TFT-W only)
40	`DISP`	1	Display and backlight enable (Neuro N6 CONN_D)

Table 20. 40-pin receptacle pin assignment. On the standard variant the SDIO and wake pins are not connected.

8.11 30-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	LCD_HSYNC	29	LTDC horizontal sync
2	SAI1_MCLK_B	30	SAI1 master clock (not used on this board)
3	BLE_TX	27	BLE UART host transmit (TFT-W only)
4	LCD_R5	28	LTDC red data bit 5
5	BLE_RX	25	BLE UART host receive (TFT-W only)
6	SCREEN_INT	26	Touch interrupt to the host (Neuro N6 CONN_N)
7	SPK_EN	23	Amplifier enable (Neuro N6 CONN_M)
8	BLE_CTS	24	BLE UART host clear to send (TFT-W only)
9	BLE_RTS	21	BLE UART host request to send (TFT-W only)
10	DEBUG_CLK	22	SWD clock (not used on this board)
11	LCD_DE	19	LTDC data enable
12	DEBUG_DAT	20	SWD data (not used on this board)
13	VIN	17	VIN rail from the Neuro N6, supplies the backlight converter
14	LCD_G4	18	LTDC green data bit 4
15	VIN	15	VIN rail from the Neuro N6, supplies the backlight converter
16	SAI1_SCK_B	16	I2S bit clock to the amplifier
17	VIN	13	VIN rail from the Neuro N6, supplies the backlight converter
18	SAI1_SD_B	14	I2S data to the amplifier
19	VIN	11	VIN rail from the Neuro N6, supplies the backlight converter
20	CONN_K	12	Wi-Fi wake-up, host to module (TFT-W only)
21	VIN	9	VIN rail from the Neuro N6, supplies the backlight converter
22	CONN_B	10	Module power-down, PDn (TFT-W only)
23	3V3	7	3.3 V supply input
24	GND	8	Ground
25	3V3	5	3.3 V supply input
26	LCD_G2	6	LTDC green data bit 2
27	1V8	3	1.8 V supply input
28	SAI1_FS_B	4	I2S frame clock to the amplifier
29	NRST	1	System reset input (camera and touch controller reset, RESET button)
30	CONN_A	2	Wi-Fi wake-up, module to host (TFT-W only)

Table 21. 30-pin receptacle pin assignment. On the standard variant the BLE UART, wake and power-down pins are not connected.

9 Laser safety

The Neuro Vision TFT and TFT-W contain a laser emitter. The VL53L1CB Time-of-Flight sensor uses an integrated vertical-cavity surface-emitting laser (VCSEL) and emits invisible 940 nm infrared light through the optical window on the top face of the sensor package while a ranging measurement is in progress. No other component on the board emits laser radiation.

9.1 Laser classification

The VL53L1CB is a Class 1 laser product and is safe under all conditions of reasonably foreseeable normal use. No protective eyewear or other protective measures are required, provided the board is used as described in this document.

Laser source	VCSEL integrated within the VL53L1CB
Laser class	Class 1
Classification standard	IEC 60825-1:2014 (Edition 3.0)
Wavelength	940 nm, invisible infrared
Field of illumination	27° full field of view
Aperture	Optical window on the top face of the VL53L1CB package



Figure 6. Class 1 laser product label, as printed on the leaflet supplied with the board.

9.2 Handling precautions

- The emitted light is invisible. Do not assume the emitter is inactive because no light can be seen: the VCSEL is enabled whenever the host has started a ranging measurement.
- Do not examine the sensor window at close range through magnifying optics such as an eye loupe, a magnifier or a microscope.
- Do not fit additional lenses, mirrors or other optical elements in front of the sensor. External optics can concentrate the beam and invalidate the Class 1 classification.
- Do not operate a board on which the VL53L1CB package or its optical window is cracked, scratched, contaminated or otherwise damaged.
- Do not rework, remove or replace the VL53L1CB. The sensor contains no user-serviceable parts, and disassembly may expose the emitter without its optical window.
- Configure the sensor through the ST VL53L1CB API or the Neuro N6 Arduino core drivers, and keep ranging parameters within the ranges given in the ST documentation.

The white illumination LED is not a laser product and falls outside the scope of IEC 60825-1. It is nevertheless a high-brightness source: avoid looking directly into the LED at close range while it is switched on.

Where the Neuro Vision TFT or TFT-W is built into an end product, the manufacturer of that product is responsible for its compliance with IEC 60825-1, including any labelling and user information.

10 Regulatory compliance

This section describes the regulatory status of the Neuro Vision TFT and Neuro Vision TFT-W as supplied by Ohm Lab. The board is an expansion board intended to be operated only when mounted on a Neuro N6, and it is assessed as part of that combination rather than as a stand-alone product. Where the board is incorporated into a finished product, responsibility for the compliance of that product rests with its manufacturer.

10.1 Product marking

SYMBOL	MARKING	APPLIED TO	MEANING
CE	CE	Board and packaging	Conformity with the EMC Directive 2014/30/EU (TFT) or the Radio Equipment Directive 2014/53/EU (TFT-W), and with the RoHS Directive 2011/65/EU.
	Crossed-out wheeled bin	Board and packaging	Do not dispose of with unsorted household waste. The bar identifies equipment placed on the market after 13 August 2005.
	Triman and Info-tri	Packaging	France: sorting information required by the French environment code. The Triman is printed on the carton; an Info-tri sticker with the sorting instructions is added to orders for France.
	Class 1 laser product	Leaflet	Explanatory label required by IEC 60825-1:2014 for the VL53L1CB emitter, printed on the leaflet supplied with the board. See the laser safety section.

Table 22. Product markings. The controlled artwork is held in the technical file.

No UKCA marking is applied: CE marking is accepted for goods placed on the Great Britain market. RoHS conformity is declared through the CE marking and carries no separate mark.

10.2 Electromagnetic compatibility

The Neuro N6 platform, including the Neuro Vision TFT and TFT-W, has been assessed for electromagnetic compatibility as a product family, based on tests of worst-case configurations of the Neuro N6 with expansion boards fitted. The Neuro Vision TFT conforms to the EMC Directive 2014/30/EU, with emissions assessed to EN 55032 Class B and immunity to EN 55035. The TFT-W is radio equipment, so its electromagnetic compatibility is covered by the Radio Equipment Directive 2014/53/EU. The radio information required by that directive is supplied with the board. The assessment applies to the board mounted on a Neuro N6 and operated with the default clock and radio settings of the Neuro N6 Arduino core. The declarations of conformity are published at neuron6.com/conformity.

10.3 Radio (TFT-W)

The Neuro Vision TFT-W contains a u-blox MAYA-W476 module providing 2.4 GHz Wi-Fi (802.11b/g/n) and BLE. The standard Neuro Vision TFT contains no radio transmitter.

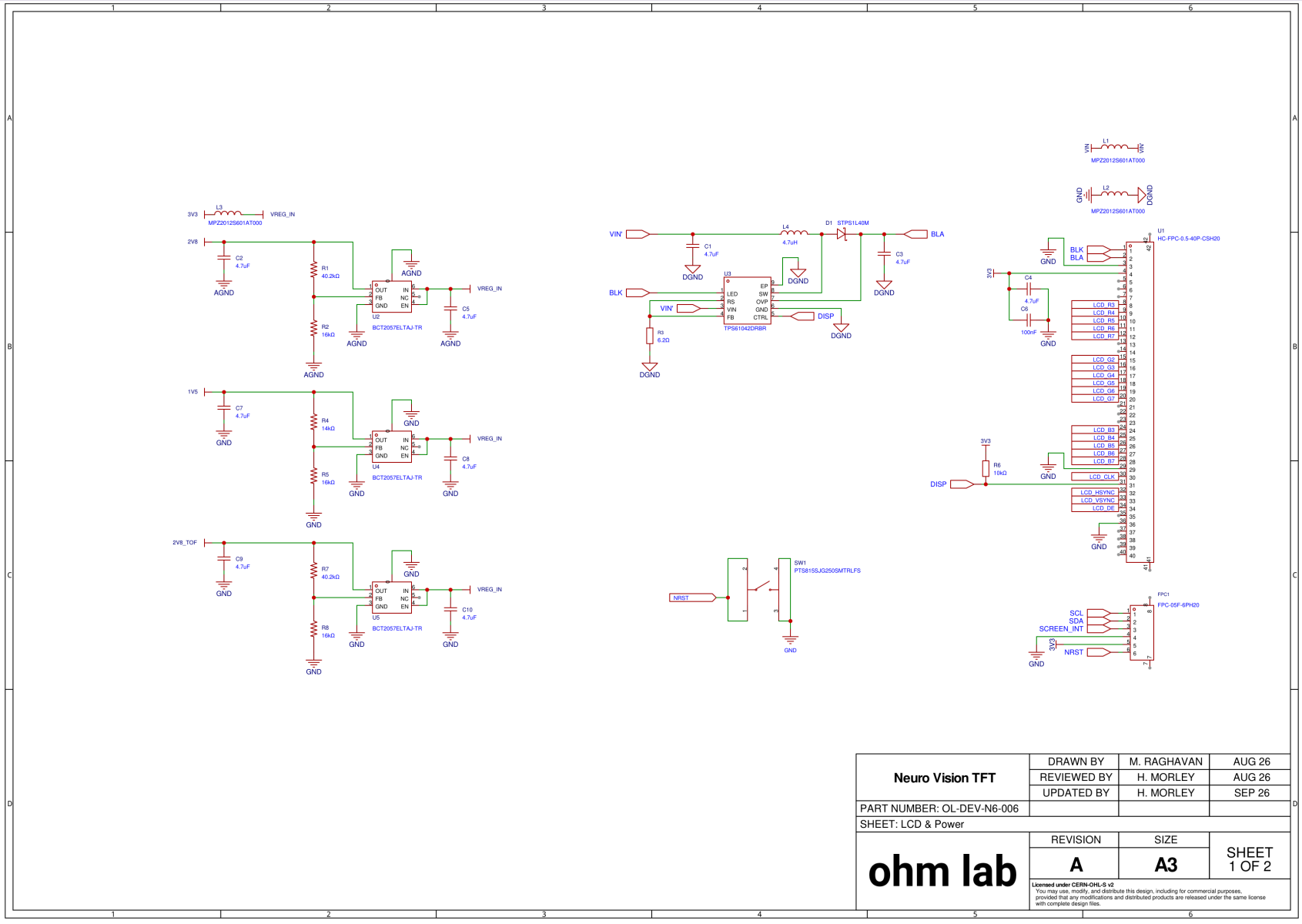
10.4 Materials and disposal

Both variants of the Neuro Vision TFT are manufactured to the substance restrictions of Directive 2011/65/EU (RoHS 2) as amended and the UK Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012. Supplier material declarations, and declarations covering substances of very high concern under REACH, are held in the technical file.

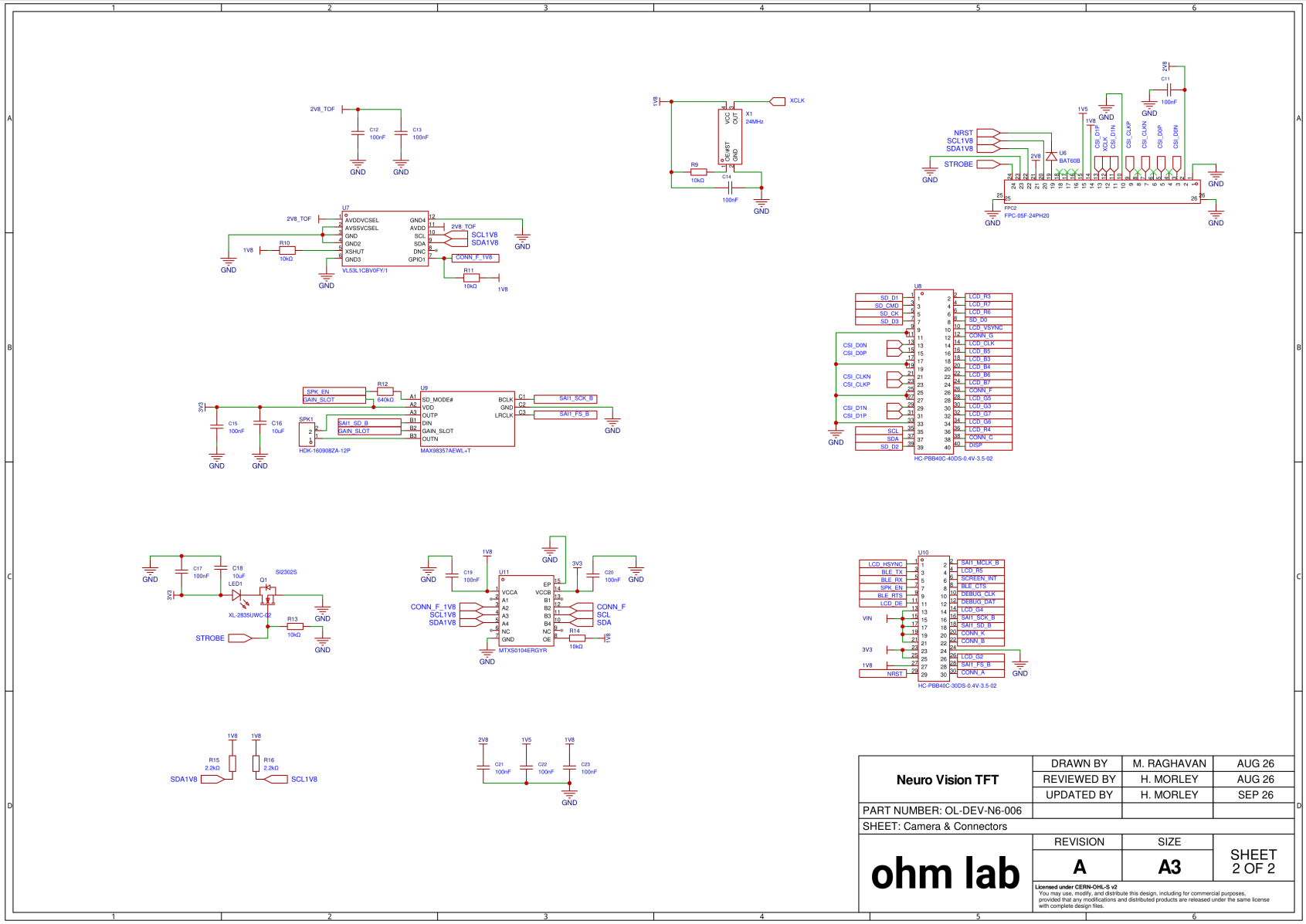
At end of life, do not dispose of the board with unsorted household waste. Return it to a designated collection point for electrical and electronic equipment, or to Ohm Lab.

11 Schematics

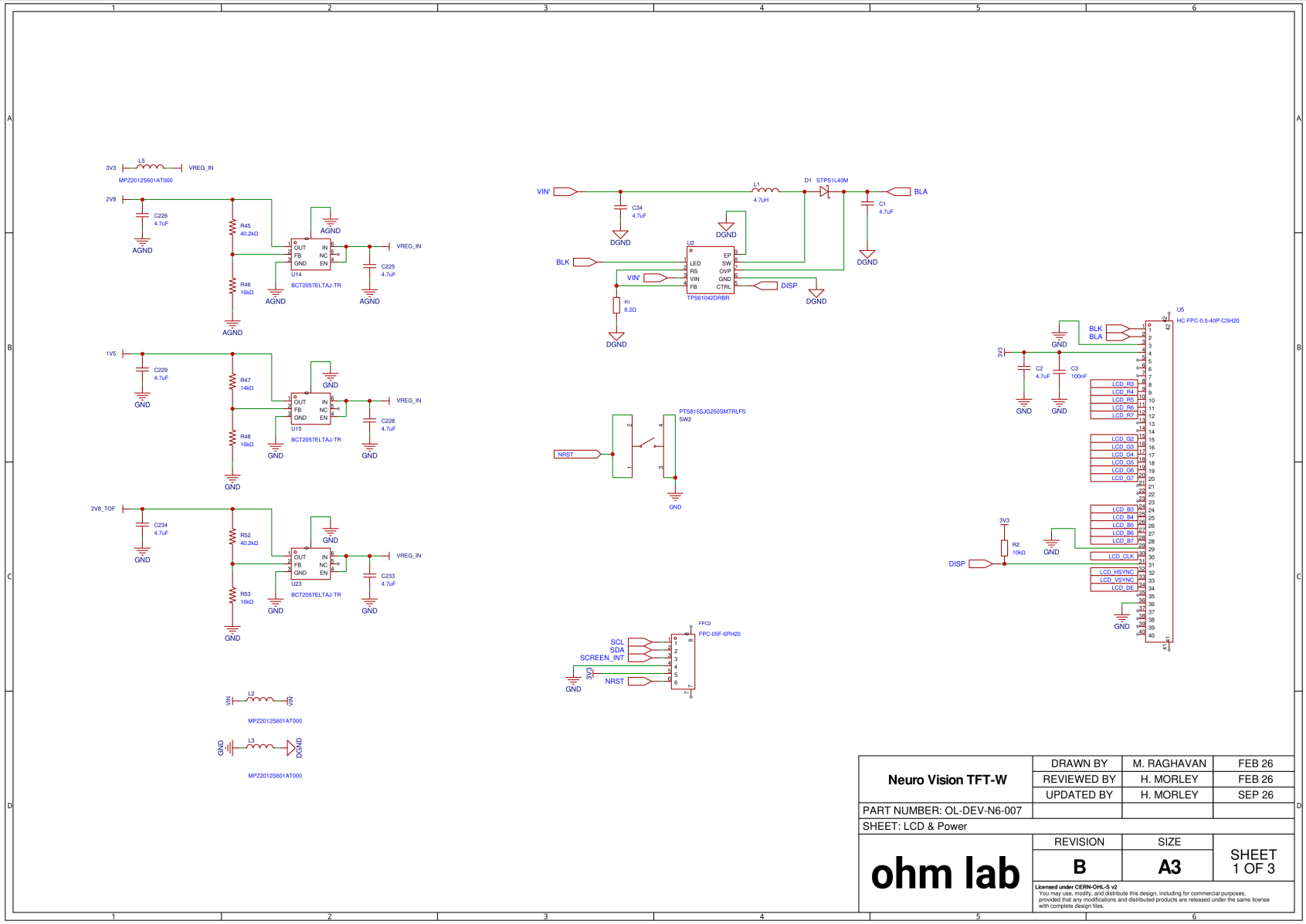
The schematics are reproduced on the following pages for reference. Net names match those used in this document; component designators differ between the two variants. Full-resolution sheets are published in the Neuro N6 Hardware repository (github.com/ohmlab-ltd/Neuro-N6-Hardware).



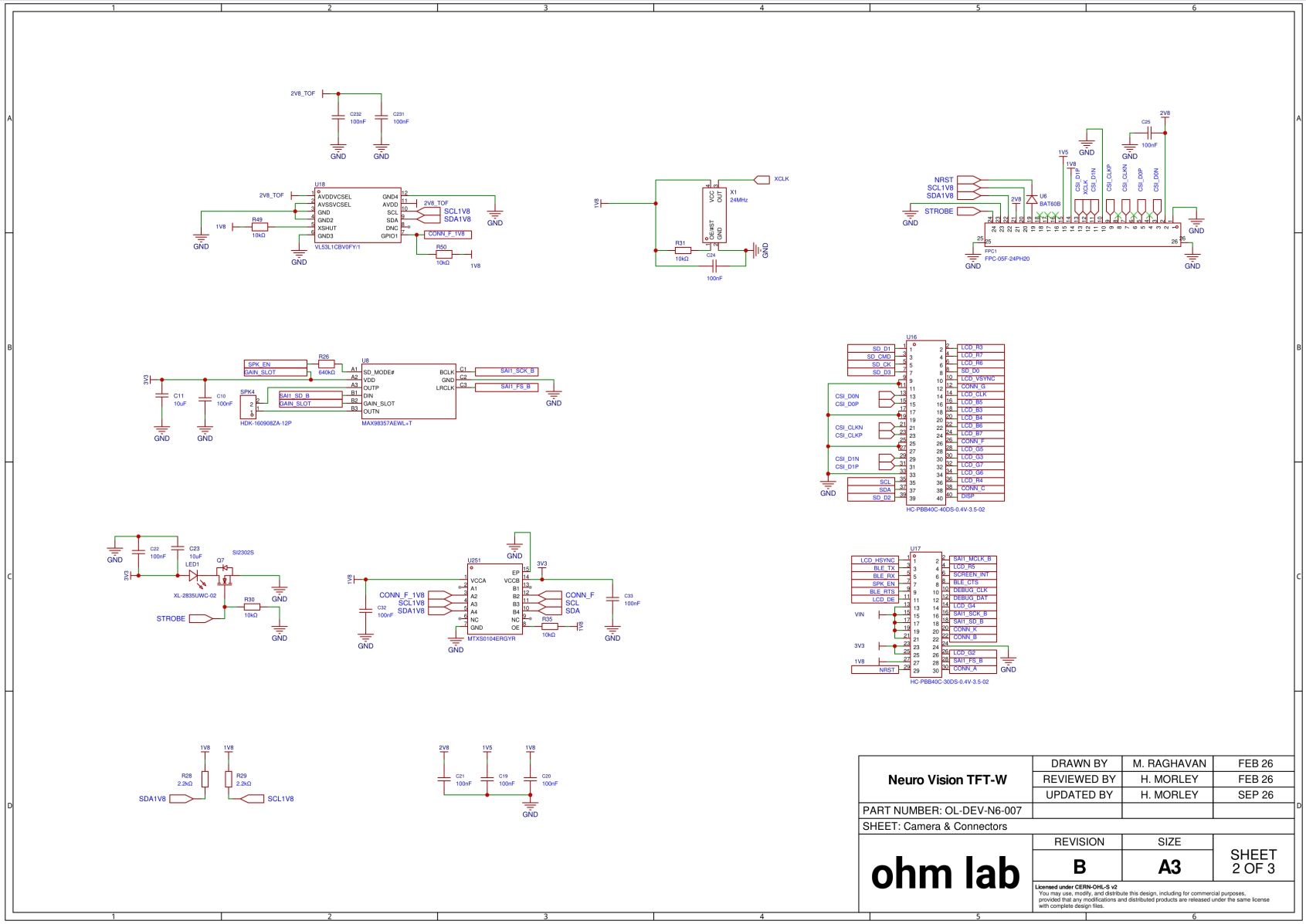
Neuro Vision TFT Rev A schematic, sheet 1 of 2: display, backlight driver, touch connector, reset button and power.



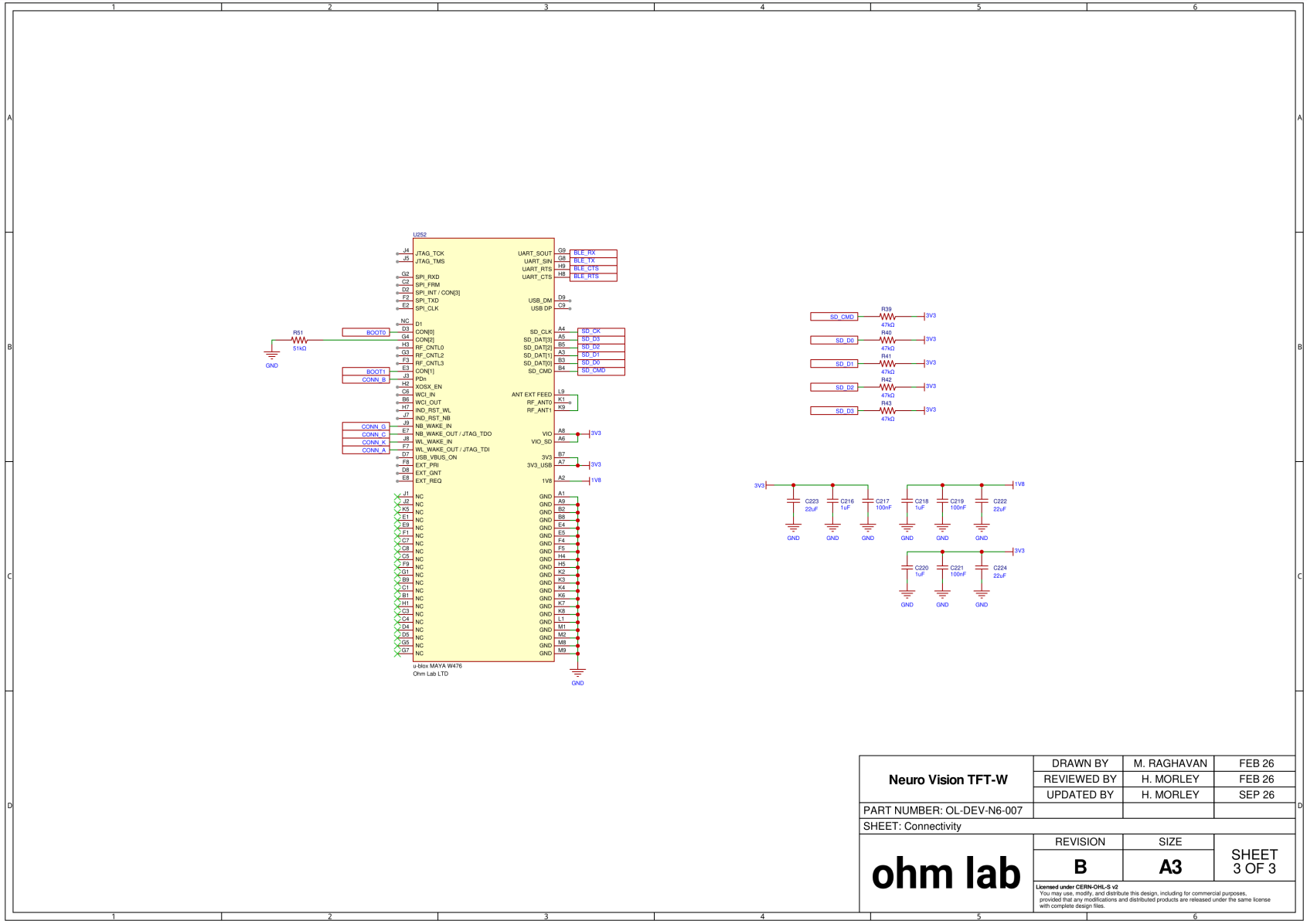
Neuro Vision TFT Rev A schematic, sheet 2 of 2: camera, Time-of-Flight sensor, illumination, audio and board-to-board connectors.



Neuro Vision TFT-W Rev B schematic, sheet 1 of 3: display, backlight driver, touch connector, reset button and power.



Neuro Vision TFT-W Rev B schematic, sheet 2 of 3: camera, Time-of-Flight sensor, illumination, audio and board-to-board connectors.



Neuro Vision TFT-W Rev B schematic, sheet 3 of 3: connectivity (MAYA-W476 module).

12 Document revision history

REVISION	DATE	CHANGES
A	02/07/2026	• Pre-production first release
B	02/09/2026	• Ratings, laser safety and regulatory section added
C	24/09/2026	• Covers the TFT and TFT-W • New block diagram and drawings
D	25/10/2026	• Features as supported by the Arduino core • Wi-Fi 2.4 GHz only

Table 23. Document revision history.